Hybrid SAR-ISAR Image Formation via Joint FrFT-WVD Processing for BFSAR Ship Target High-Resolution Imaging

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***Abstract*— Bistatic forward-looking synthetic aperture radar (BFSAR) is a kind of bistatic SAR system that can image forward-looking terrain in the flight direction of the receiver. Current literature and reports about BFSAR mainly concentrate on the stationary scene and ground-moving target imaging. Unlike stationary and ground-moving targets, the translational and rotational movements of ship targets usually lead to complicated range cell migration (RCM) and Doppler frequency migration (DFM). Moreover, the characteristics of RCM and DFM for different scattering points of the ship target are significantly different, i.e., the characteristics of the RCM and DFM are 2-D spatial variation, ultimately leading to severe defocusing of ship target in the SAR image. To solve these problems, a kind of hybrid SAR-ISAR imaging formation is proposed for BFSAR ship target imaging. First, to solve the problem of the Doppler ambiguity caused by the forward-looking mode of the receiver, an efficient ambiguity estimation method based on the minimum entropy criterion is presented. Then, keystone transform and range alignment processing can be applied to correct the spatial variant range walk and higher order RCM, respectively. Moreover, in order to obtain a high-resolution and well-focused image after translational compensation, a new method based on the fractional Fourier transform (FrFT) and the Wigner–Ville distribution (WVD) is proposed, where FrFT is applied to separate the multiple main scattering points in each range cell, and WVD is applied to obtain the high-resolution time–frequency distribution of each scattering point. Compared with the conventional ISAR range-Doppler (RD) algorithm and time–frequency estimation-based imaging methods, this method not only has no cross terms but also has high processing accuracy and better antinoise performance.**

***Index Terms*— Bistatic forward-looking synthetic aperture radar (BFSAR), fractional Fourier transform (FrFT), inverse synthetic aperture radar (ISAR), ship target, Wigner–Ville distribution (WVD).**

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I. Introduction

SYNTHETIC aperture radar (SAR) has been widely used in automatic guidance, resource detection, military reconnaissance, and so on as it can obtain high-resolution images of ground objects independent of time and weather [1]–[3]. However, monostatic radar cannot achieve forward-looking area imaging due to the limitation of the imaging mechanism [4], [5]. With the development of SAR technology, bistatic SAR has been proposed. Since the radar receiver and transmitter are placed on different platforms, bistatic SAR has advantages of concealment, stronger antijamming ability, and more abundant characteristics compared with monostatic SAR. Besides, by properly adjusting the beam direction of the radar, the bistatic SAR can achieve the imaging of the forward-looking region of the receiver, which is called bistatic forward-looking SAR (BFSAR) [6]–[9]. With the development of the BFSAR system, it is also necessary to focus on moving targets in the scene, such as ship targets on the sea, due to the importance of moving target information to the surveillance system [10].

Unlike the stationary and ground-moving targets, the translational and rotational movements of ship targets usually lead to complicated range cell migration (RCM) and Doppler frequency migration (DFM). Moreover, the characteristics of RCM and DFM for different scattering points of the ship target are significantly different, i.e., the characteristics of the RCM and DFM are 2-D spatial variation, which ultimately leads to severe defocusing of ship target in the SAR image. What is worse is that the ship target performs an unsteady 3-D swing under the influence of sea waves, and the ship target imaging becomes much more complex and difficult in BFSAR.

Nevertheless, current literature and reports of BFSAR mainly focus on researching the imaging algorithms for stationary scenario, such as Omega-k algorithms [11], [12], nonlinear chirp scaling (NLCS) algorithms [13], [14], and range-Doppler (RD) algorithms [15]. However, these methods cannot be applied to ship target imaging due to the additional Doppler modulation and the complicated RCM caused by the noncooperative movement of the ship target. At present, some ground-moving target imaging (GMTI) methods based on the Doppler parameter compensation are proposed [16], [17].

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However, for the Doppler parameter estimation methods, such as the time-chirp rate distribution method [18] and the time–frequency analysis method [19], cross terms emerge since there exists more than one scattering point in one range cell. To make matters worse, since the Doppler parameter of BFSAR is a spatial variant, it is not appropriate to use the same estimated Doppler parameter to compensate for the whole scattering points of the ship target. In general, none of the aforesaid methods can be applied to BFSAR ship target imaging. For the spatial variant DFM equalization problem, a spatial-dechirp function is designed to eliminate the spatial frequency extension [13], [14], and NLCS is applied to equalize the spatial-variant Doppler parameters of stationary scenario, but it is not applicable to noncooperative moving targets. Based on motion parameter estimation, some monostatic GMTI methods in [20]–[22] and some BFSAR GMTI methods, such as mismatched compression method [23], spectrum matching method [24], and keystone-based delay-correlation transform and fast searching Fourier transform (KDCT-FSFT) refocusing method [25], are proposed. However, only the translational movement is considered in these methods, which is unsuitable for ship target imaging.

Moreover, due to the complexity of rotational movement and the timely variation of Doppler parameters of ship targets, the traditional inverse SAR (ISAR) RD algorithm generates a severely defocused result of it. To address this issue, an effective range-instantaneous-Doppler (RID) technique is proposed [26], [27], in which the cross-range resolution is provided by the instantaneous Doppler frequency. Existing nonparametric estimation methods, such as the Wigner–Ville distribution (WVD) [28]–[30] and the short-time Fourier transform (STFT), are computationally efficient, but none of these methods can be directly applied to high-precision imaging due to serious cross terms in WVD and mutually restricted resolution between time and frequency in STFT. Some parametric-estimation-based RID imaging methods have also been proposed in [31]–[35], where the target signal of each range gate has been modeled as a multicomponent linear frequency modulation (LFM) signal or multicomponent cubic phase signal (CPS) after translation compensation. These parametric-estimation-based reconstruction methods, such as the adaptive chirplet-decomposition [31], [32], the cubic phase function (CPF) [33]–[35], and the scaled Fourier transform (SCFT) [34], [36], [37], have a heavy computation burden. At the same time, because of the nonlinearity of the kernel function, cross terms will inevitably be introduced, which affects the estimation accuracy of the multicomponent parameters.

In this article, a hybrid SAR-ISAR image formation via joint fractional Fourier transform (FrFT)-WVD processing is proposed for ship targets in BFSAR. First, to solve the problem of the Doppler ambiguity caused by the forward-looking mode of the receiver, an efficient ambiguity estimation method based on the minimum entropy criterion is presented. Then, according to the estimated parameter associated with the Doppler centroid, the prefilter function is constructed to eliminate the Doppler ambiguity. Third, keystone transform (KT) and range alignment processing can be applied to correct the spatial variant range walk and spatial invariant higher order RCM, respectively. Finally, after the translational compensation, joint FrFT-WVD processing is carried out for each range gate to obtain a well-focused ISAR image. The advantages of this method are not only the absence of cross terms but also the high processing accuracy and better antinoise performance.

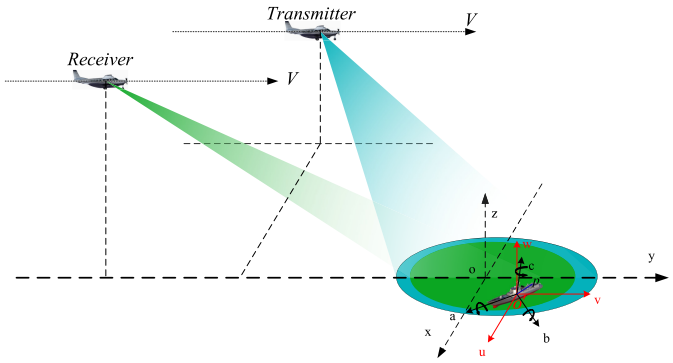


Fig. 1. BFSAR imaging geometric configuration of ship target.

The remainder of this article is organized as follows. Section II gives the imaging geometric configuration, signal model of ship target in BFSAR, and also with signal chrematistic analysis. The joint FrFT-WVD image formation is introduced in Section III, and the main steps of this imaging method are given in Section IV. The effectiveness of the proposed method is proved by simulation results in Section V. Finally, the conclusion of this article is given in Section VI.

II. Imaging Geometry and Signal Model

In this section, the BFSAR imaging geometrical configuration for the ship target is constructed first, and then, the signal model is given. At the same time, the properties of RCM and Doppler parameters in BFSAR echo are analyzed in detail.

A. Geometric Configuration

The 3-D coordinate systems include the global coordinates XYZ, and the local coordinates UVW are depicted in Fig. 1. Locations of the points on the ship are defined in the ship abc coordinates. The positions of the BFSAR receiver and transmitter in the XYZ coordinates system are (X_R, Y_R, H_R), (X_T, Y_T, H_T), respectively. Supposing that the transmitter and receiver fly forward along the Y-axis at the speed of V.

The ship target moves in a translational motion with a velocity of $[v_x, v_y, 0]^T$. At the same time, it performs 3-D swings under the influence of the sea waves. The 3-D degrees of swing include roll, pitch, and yaw, respectively. According to ship dynamics, it can be described as follows:

$$\theta_i(\eta) = \frac{1}{2} A_i \cos\left(2\pi \frac{\eta}{T_i} + \psi_i\right), \quad i = \text{roll, pitch, yaw} \quad (1)$$

where A_i represents the swing amplitude, T_i represents the swing period, and ψ_i represents the initial phase of the swing. The position of a scatterer point p on the ship target in the abc coordinates system is $[a_p\ b_p\ c_p]^T$. After the hull has been shaken in three dimensions, the location of the point p in the

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UVW coordinates system can be expressed as

$$\mathbf{r} = \left[u_p, v_p, w_p\right]^T = \Re(\eta)\left[a_p, b_p, c_p\right]^T \tag{2}$$

$$\Re(\eta) = \Re_{\text{roll}}(\eta)\Re_{\text{pitch}}(\eta)\Re_{\text{yaw}}(\eta) \tag{3}$$

where $\Re_{\text{roll}}(\eta)$, $\Re_{\text{pitch}}(\eta)$, and $\Re_{\text{yaw}}(\eta)$ are the roll, pitch, and yaw angles describing the rotation around the a-, b-, and c-axes, respectively. These can be expressed by

$$\Re_{\text{roll}}(\eta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_{\text{roll}}(\eta) & -\sin\theta_{\text{roll}}(\eta) \\ 0 & \sin\theta_{\text{roll}}(\eta) & \cos\theta_{\text{roll}}(\eta) \end{bmatrix} \tag{4}$$

$$\Re_{\text{pitch}}(\eta) = \begin{bmatrix} \cos\theta_{\text{pitch}}(\eta) & 0 & \sin\theta_{\text{pitch}}(\eta) \\ 0 & 1 & 0 \\ -\sin\theta_{\text{pitch}}(\eta) & 0 & \cos\theta_{\text{pitch}}(\eta) \end{bmatrix} \tag{5}$$

$$\Re_{\text{yaw}}(\eta) = \begin{bmatrix} \cos\theta_{\text{yaw}}(\eta) & -\sin\theta_{\text{yaw}}(\eta) & 0 \\ \sin\theta_{\text{yaw}}(\eta) & \cos\theta_{\text{yaw}}(\eta) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{6}$$

Supposing the initial value of the point o' (the coordinates origin of UVW) in the XYZ coordinates system is $[u_0\ v_0\ w_0]^T$. After the ship moves for a period of time, the location value of the point p in the XYZ coordinates system can be expressed as follows:

$$\begin{bmatrix} x_p(\eta) \\ y_p(\eta) \\ z_p(\eta) \end{bmatrix} = \Re(\eta)\begin{bmatrix} a_p \\ b_p \\ c_p \end{bmatrix} + \begin{bmatrix} u_0 \\ v_0 \\ w_0 \end{bmatrix} + \begin{bmatrix} v_x \\ v_y \\ 0 \end{bmatrix}\eta. \tag{7}$$

B. Signal Model

Assuming that an LFM signal is transmitted, i.e.,

$$s(t) = \text{rect}\left(\frac{t}{T_p}\right)\exp\left[j2\pi\left(f_c t + \frac{1}{2}K_r t^2\right)\right] \tag{8}$$

where T_p, f_c, and K_r denote the pulse duration, carrier frequency, and frequency modulated rate, respectively. The received echo after demodulation can be expressed as

$$s(\eta, t) = \text{rect}\left(\frac{\eta}{T_a}\right)\text{rect}\left(\frac{t - R(\eta)/c}{T_p}\right) \times \exp\left\{j\pi K_r\left(t - \frac{R(\eta)}{c}\right)^2\right\}\exp\left\{-\frac{j2\pi f_c R(\eta)}{c}\right\} \tag{9}$$

where η is the slow time, t is the fast time, T_a denotes the aperture time, and c is the speed of electromagnetic waves. $R(\eta)$ is the instantaneous bistatic range, and it can be given by

$$R(\eta) = R_T(\eta) + R_R(\eta) \tag{10}$$

where $R_T(\eta)$ and $R_R(\eta)$ are point p relative to the distance history of the receiver and transmitter, respectively. $R_T(\eta)$ and $R_R(\eta)$ are given in (12) and (11), as shown at the bottom of the page. Considering that the ship moves along an unknown trajectory, and expanding $R(\eta)$ at the beam center crossing time $\eta = 0$ into the Maclaurin series, we have

$$R(\eta) = R(0) - \lambda f_{dc}\eta - \frac{1}{2}\lambda f_{dr}\eta^2 \cdots \tag{15}$$

where

$$\begin{cases} f_{\text{dc}} = -\dfrac{1}{\lambda}\dfrac{dR(\eta)}{d\eta}\Big|_{\eta=0} \\ f_{\text{dr}} = -\dfrac{1}{\lambda}\dfrac{d^2R(\eta)}{d\eta^2}\Big|_{\eta=0}. \end{cases} \tag{16}$$

In general, to simplify the analysis, the range history in (10) is considered as a combination of translational and rotational components in the turntable model of ISAR imaging. Usually, the size of the ship is much smaller than the distance between the platforms and the ship target. Therefore, formulas (11) and (12) can be rewritten, and the following approximations can be made:

$$R_T(\eta) = \|R_{\text{To}'}(\eta)\mathbf{n_T}(\eta) + \mathbf{r}(\eta)\| \approx R_{\text{To}'}(\eta) + \mathbf{n_T}(\eta)^T\mathbf{r}(\eta) \tag{17}$$

$$R_R(\eta) = \|R_{\text{Ro}'}(\eta)\mathbf{n_R}(\eta) + \mathbf{r}(\eta)\| \approx R_{\text{Ro}'}(\eta) + \mathbf{n_R}(\eta)^T\mathbf{r}(\eta). \tag{18}$$

In the XYZ coordinate system, $\mathbf{n_R}(\eta)$ and $\mathbf{n_T}(\eta)$ represent the unit vector of the receiver and transmitter lines of sight, respectively. $\mathbf{r}(\eta)$ is the vector representation of the scattering point p in the UVW coordinates system. $R_{\text{Ro}'}(\eta)$ and $R_{\text{To}'}(\eta)$, respectively, represent the distance between the receiver and transmitter to the center of the ship. $R_{\text{Ro}'}(\eta)$ and $R_{\text{To}'}(\eta)$ are given in (13) and (14), as shown at the bottom of the page

$$R(\eta) = R_{\text{tr}}(\eta) + R_{\text{ro}}(\eta). \tag{19}$$

The translation component and the rotational component can be given by

$$R_{\text{tr}}(\eta) = R_{\text{Ro}'}(\eta) + R_{\text{To}'}(\eta) \tag{20}$$

$$R_{\text{ro}}(\eta) = (\mathbf{n_T}(\eta) + \mathbf{n_R}(\eta))^T\mathbf{r}(\eta) = (\mathbf{n_T}(\eta) + \mathbf{n_R}(\eta))^T\Re(\eta)\left[a_p, b_p, c_p\right]^T. \tag{21}$$

In order to simplify the analysis, the translation component is expanded by the following polynomial:

$$R_{\text{tr}}(\eta) = a_0 + a_1\eta + a_2\eta^2 + a_3\eta^3 \cdots \tag{22}$$

$$R_R(\eta) = \sqrt{\left(X_R - x_p(\eta)\right)^2 + \left(Y_R + V\eta - y_p(\eta)\right)^2 + \left(H_R - z_p(\eta)\right)^2} \tag{11}$$

$$R_T(\eta) = \sqrt{\left(X_T - x_p(\eta)\right)^2 + \left(Y_T + V\eta - y_p(\eta)\right)^2 + \left(H_T - z_p(\eta)\right)^2} \tag{12}$$

$$R_{Ro'}(\eta) = \sqrt{(X_R - u_0 - v_x\eta)^2 + \left(Y_R + V\eta - v_0 - v_y\eta\right)^2 + (H_R - w_0)^2} \tag{13}$$

$$R_{To'}(\eta) = \sqrt{(X_T - u_0 - v_x\eta)^2 + \left(Y_T + V\eta - v_0 - v_y\eta\right)^2 + (H_T - w_0)^2} \tag{14}$$

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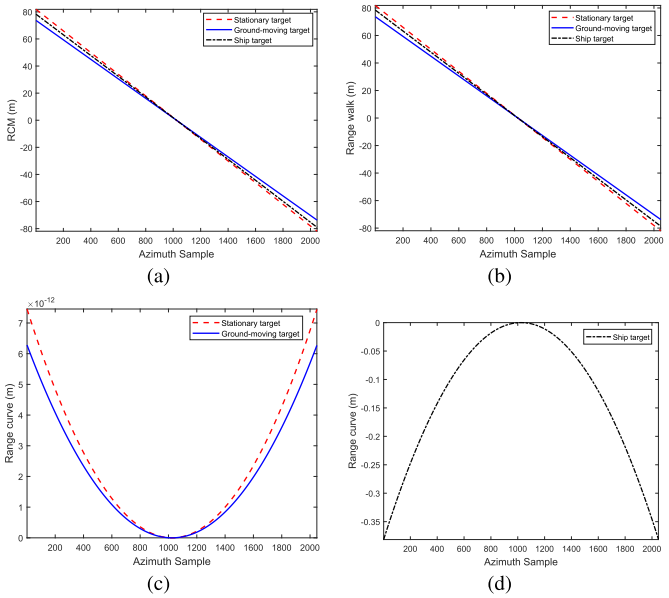


Fig. 2. RCM comparison of different kinds of target. (a) RCM. (b) Range walk. (c) Range curve of ground-moving target and stationary target. (d) Range curve of ship target.

Similarly, the unit vector of the line of sight of the receiver and transmitter can be expanded by the Taylor series

$$\begin{aligned}\mathbf{n_R}(\eta) &= \mathbf{n_R}(0) + \mathbf{n'_R}(0)\eta + \frac{1}{2}\mathbf{n''_R}(0)\eta^2 \cdots \\ \mathbf{n_T}(\eta) &= \mathbf{n_T}(0) + \mathbf{n'_T}(0)\eta + \frac{1}{2}\mathbf{n''_T}(0)\eta^2 \cdots \end{aligned} \tag{23}$$

whereas the term $((\mathbf{n'_R}(0) + \mathbf{n'_T}(0))\eta)^T \Re(\eta)[a_p, b_p, c_p]^T$ corresponds to the range and azimuth positions. It is also noted that this term may induce RCM. Fortunately, the effect of liner RCM can be effectively eliminated by KT [38] in a low SNR condition.

Up until now, the signal model of the ship target in BFSAR has been constructed. In Section II-C, the characteristics of the ship target in BFSAR are analyzed in detail.

C. Echo Characteristic Analysis

In this section, the properties of BFSAR echo are analyzed in detail. The BFSAR system parameters are the same as those in Table II, and the rotational parameters are given in Table III. The point target of the simulation analysis is located in (50 m, 50 m, 0 m) with a velocity of (7 m/s, 11 m/s, 0 m/s). In order to better analyze the echo characteristics, the stationary target and the ground-moving target (i.e., the target only with translation motion) are also analyzed.

From (15), RCM of the ship target with an unknown motion in BFSAR can be expressed as

$$\begin{aligned} R_{\text{rcm}}(\eta) &= b_1\eta + \frac{1}{2}b_2\eta^2 \\ &= -\lambda\left(f_{dc}\eta + \frac{1}{2}f_{dr}\eta^2\right). \end{aligned} \tag{24}$$

Inspecting (24), the first term is related to f_{dc}, and the second term is related to f_{dr}, respectively. The analysis results of RCM are shown in Fig. 2. The range walk and the range curve of these three targets are given in Fig. 2(b)–(d). The red lines represent those of the stationary target, the blue lines represent the ground-moving target, and the black lines represent the ship target with both translation and rotation motions. From Fig. 2(a), it can be observed that range walk is the major part of the RCMs in the BFSAR. Due to the unknown motion, we can observe that all the RCMs of the ground-moving target and the ship target with both translation and rotation motions are different from the stationary target. Fig. 2(c) and (d) shows that the maximum range curvature of the ship target with both translation and rotation motions is larger than the ground-moving target and stationary target, and even the curvature direction changes.

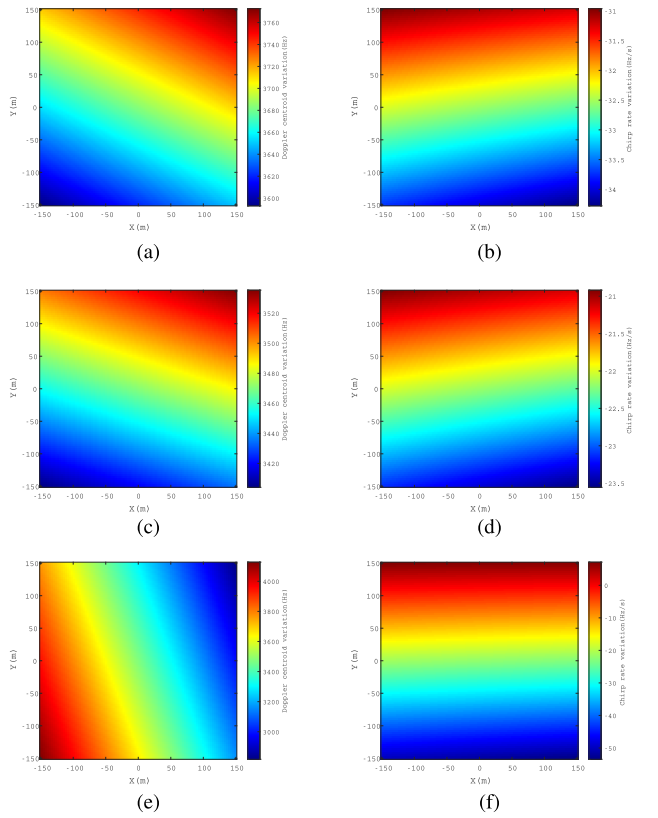


Fig. 3. Doppler property of different kinds of target. (a) Doppler centroid property of stationary target. (b) Chirp rate property of stationary target. (c) Doppler centroid property of ground-moving target. (d) Chirp rate property of ground-moving target. (e) Doppler centroid property of ship target. (f) Chirp rate property of ship target.

Considering that the size of the ship target is usually smaller than 150 m, in order to analyze the spatial variant, Doppler parameters in an imaging scene of about 90 000 m^2 in the X-direction $(-150\text{ m}, 150\text{ m})$ and the Y-direction $(-150\text{ m}, 150\text{ m})$ are analyzed. The analysis results of the Doppler property are shown in Fig. 3. Fig. 3(a), (c), and (e) shows the variation of f_{dc} related to b_1 with different positions. The color scale is in Hz. It can be noticed that f_{dc} of the ship target ranges from about 3000 to 4000 Hz, which is much larger than the pulse repetition frequency (PRF) of the BFSAR system, and its DFM is larger than a stationary target. As a consequence, its Doppler spectrum is more easily split into two adjacent PRF bands, which is called the Doppler ambiguity

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phenomenon. This Doppler ambiguity can cause the KT to be unable to correct the range walk. Thus, before the range walk correction through the KT, the effect of the Doppler centroid ambiguity must be eliminated.

Fig. 3(b), (d), and (f) shows the variation of f_{dr} related to b_2 with different positions. The color scale is in Hz/s. It can be seen from these three figures that f_{dr} of ground-moving target and ship target is different from the stationary target. As a result, if the ship target is compensated with the Doppler parameter of the stationary target, the image will be severely defocused. At the same time, it can be found that the chirp rate is the 2-D spatial variant. Thus, if all the range cells are compensated with the same Doppler parameter, the image will also be defocused. Moreover, the defocus will be more serious due to the larger chirp rate migration compared with stationary target and ground-moving target.

According to the echo characteristic analysis, three main problems for high precision imaging can be concluded: 1) RCM is not only very large but also changes with the spatial location of scattering point; 2) the large Doppler frequency causes Doppler centroid ambiguity in BFAR; and 3) 2-D spatial variant chirp rate of ship target makes it difficult to obtain a well-focused image. Section III will focus on how to deal with the aforesaid three issues.

III. Proposed Method

A. Doppler Ambiguity Estimation

Due to the forward-looking mode of the receiver, the received echo of the BFSAR often has a large Doppler frequency, which causes Doppler centroid ambiguity. After the BFSAR and the ship's motion are equivalent to the turntable model, it can be considered that the Doppler ambiguity is mainly caused by the larger first order in the translation polynomial. The first-order coefficient is only used to estimate the Doppler ambiguity but not for phase compensation, so its estimation accuracy does not need to be too high. This article proposes a method based on the global range alignment (GRA) to estimate the Doppler ambiguity in range-compressed echo. The echo after range compression can be expressed as

$$s_1(\eta, t) = \mathrm{rect}\left(\frac{\eta}{T_a}\right) \sin c\left(t - \frac{R(\eta)}{c}\right) \exp\left\{-\frac{j2\pi f_c R(\eta)}{c}\right\}. \tag{25}$$

The average range profile (ARP) is defined as the sum of the envelope along the azimuth after range compression [39]. Let $s_2(\eta, t) = |s_1(\eta, t)|$; then, the mathematical expression of the ARP can be given by

$$s_{\mathrm{rim}}(t) = \int_{-T/2}^{T/2} s_2(\eta, t + \Delta t(\eta)) d\eta \tag{26}$$

where $\Delta t(\eta)$ is the time offset of the range gate corresponding to each azimuth time. According to the time-shift property of the Fourier transform (FT), (26) can be rewritten as

$$s_{\mathrm{rim}}(t) = \int_{-T/2}^{T/2} \int_{f_r} S_2(\eta, f_r) e^{j2\pi f_r \Delta t(\eta)} \cdot e^{j2\pi f_r t} d f_r d\eta \tag{27}$$

where $S_2(\eta, f_r) = \mathrm{FFT}[s_2(\eta, t)]$ is the result of FT along the range of the echo amplitude value after range compression and f_r is the range frequency.

The Shannon entropy can measure the focusing quality of the image. In this article, the entropy of the ARP is used to reflect the effectiveness of range alignment [39]. The entropy of the ARP is given by

$$H = -\int \rho(t) \ln \rho(t) dt \tag{28}$$

where

$$\rho(t) = \frac{s_{\mathrm{rim}}(t)}{\int s_{\mathrm{rim}}(t) dt}. \tag{29}$$

It can be observed that $\int s_{\mathrm{rim}}(t) dt$ is a constant, so the minimum of formula (28) is equivalent to the minimum of formula (30)

$$H_2 = -\int s_{\mathrm{rim}}(t) \ln s_{\mathrm{rim}}(t) dt. \tag{30}$$

When H_2 reaches the minimum value, its corresponding gradient value is 0

$$\frac{\partial H_2(t)}{\partial \Delta t(\eta)} = 0. \tag{31}$$

Let $\tilde{s}_{\mathrm{rim}}(t) = \ln s_{\mathrm{rim}}(t)$; substituting (27) and (30) into (31), one obtains

$$\int_{f_r} j2\pi f_r S_2(\eta, f_r) \tilde{S}^*_{\mathrm{rim}}(f_r) \cdot e^{j2\pi f_r \Delta t(\eta)} dt = 0 \tag{32}$$

where $\tilde{S}^*_{\mathrm{rim}}(f_r)$ is the FT of $\tilde{s}_{\mathrm{rim}}(t)$. Based on the differential properties of the FT, (32) is equivalent to finding $\Delta t(\eta)$ that satisfies the following equation:

$$\frac{d}{dt}[s_2(\eta, t) \otimes \tilde{s}_{\mathrm{rim}}(-t)] = 0. \tag{33}$$

It can be seen from (33) that the corresponding $\Delta t(\eta)$ can be obtained by iteratively solving the correlation peak positions of $s_2(\eta, t)$ and $\tilde{s}_{\mathrm{rim}}(-t)$. After all pulses are shifted correspondingly, ARP is updated so that the entropy of ARP gradually approaches the global minimum. After solving $\Delta t(\eta)$ iteratively, according to $\Delta R(\eta) = \Delta t(\eta) c = a_1\eta + a_2\eta^2 + a_3\eta^3, \ldots$, the coefficient $\hat{a}_1$ is obtained by polynomial fitting. Then, the Doppler ambiguity number of the BFSAR echo can be estimated by the above method, and it can be expressed as

$$\hat{N} = \mathrm{round}\left(\frac{\hat{a}_1}{\lambda \mathrm{PRF}}\right) \tag{34}$$

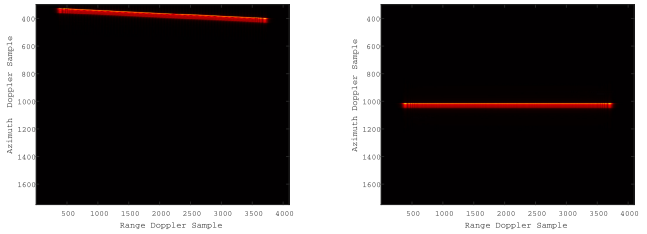
where PRF is the azimuth sampling frequency. Compared with the Radon transform [23], it does not require searching for rotation angles. The convolution operation of formula (33) can be realized quickly by FFT.

B. Doppler Ambiguity Elimination

The Doppler ambiguity may cause the echo spectrum to be split into two adjacent PRF bands [23]. Thus, a Doppler deblurring prefiltering function should be constructed to suppress Doppler ambiguity first, and the prefiltering function expression is

$$H_{\mathrm{pre}} = \exp\left\{j2\pi \frac{f_r + f_c}{c} \hat{a}_1 \eta\right\}. \tag{35}$$

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(a) (b)

Fig. 4. Echo spectrum. (a) Before prefiltering processing. (b) After prefiltering processing.

Performing the FT with respect to the fast time variable t, the received signal in the range frequency azimuth time domain can be expressed as

$$S_1(\eta, f_r) = \text{rect}\left(\frac{\eta}{T_a}\right)\text{rect}\left(\frac{f_r}{B_r}\right)\exp\left\{-j2\pi\frac{f_r+f_c}{c}R(\eta)\right\} \tag{36}$$

where B_r is the bandwidth. By multiplying (35) with (36), the prefiltered signal can be expressed as

$$S_2(\eta, f_r) = \text{rect}\left(\frac{\eta}{T_a}\right)\text{rect}\left(\frac{f_r}{B_r}\right) \times \exp\left\{-j2\pi\frac{f_r+f_c}{c}\begin{pmatrix} R_{ro}(\eta)+R_0+a_1'\eta \\ +a_2\eta^2+a_3\eta^3\ldots \end{pmatrix}\right\} \tag{37}$$

where $a_1' = a_1 - \hat{a}_1$ is the residual first-order coefficient of the translational range $R_{\text{tr}}(\eta)$.

The echo-Doppler spectrum of ship target before and after prefiltering is shown in Fig. 4. From Fig. 4(a), it can be found that the Doppler spectrum of the ship target in bistatic SAR is inclined and nonorthogonal along the 2-D frequency direction, and it is not at the center of the Doppler spectrum and is almost split into two adjacent PRF bands. From Fig. 4(b), we can find that the Doppler spectrum has been shifted to the center through prefiltering. At the same time, it is worth noting that the coupling relationship has been reduced, and the 2-D orthogonal relationship of the echo has been increased via prefiltering. It can also be seen from formula (37) that the coefficient of the first-order term has become smaller after the prefiltering process.

C. Translational Compensation

The BFSAR has a large first-order range migration, and its RCM is a spatial variant, because of the receiver's forward-looking. The traditional ISAR range alignment methods, such as correlation method, accumulation correlation method [40], and the GRA method mentioned in this article, cannot align the spatial variant envelope, but it can be easily solved through KT [38]. This article first uses the KT to correct the spatial variant range walk and then uses the GRA method to align the higher-order terms and, finally, complete the echo envelope alignment.

After the Doppler deblurring is performed on the echo, the KT is utilized to correct the range walk of the moving ship target, and it is given by

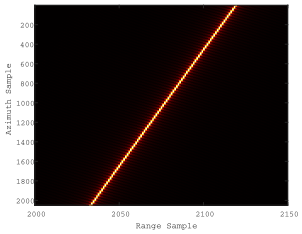


Fig. 5. Range compression result of ship target.

$$\eta = \frac{f_c}{f_c+f_r}\xi. \tag{38}$$

After the KT, the signal becomes

$$S_2(\xi, f_r) = \text{rect}\left(\frac{\xi}{T_a}\right)\text{rect}\left(\frac{f_r}{B_r}\right)\exp\{-j\Phi(\xi, f_r)\} \tag{39}$$

where

$$\Phi(\xi, f_r) = \Phi_{ro}(\xi, f_r) + \Phi_{tr}(\xi, f_r) \tag{40}$$

$$\Phi_{\text{tr}}(\xi, f_r) = \frac{2\pi(f_r+f_c)}{c}R_0 + \frac{2\pi a_1' f_c}{c}\xi + \frac{2\pi a_2 f_c}{c}\frac{f_c}{f_c+f_r}\xi^2 + \frac{2\pi a_3 f_c}{c}\left(\frac{f_c}{f_c+f_r}\right)^2\xi^3 \tag{41}$$

$$\Phi_{ro}(\xi, f_r) = \frac{2\pi(f_r+f_c)}{c}\mathbf{r}^T(\mathbf{n}_T(0)+\mathbf{n}_{\mathbf{R}}(0)) + \frac{2\pi f_c}{c}\mathbf{r}^T\left(\mathbf{n}_T'(0)+\mathbf{n}_{\mathbf{R}}'(0)\right)\xi + \frac{\pi f_c}{c}\frac{f_c}{f_c+f_r}\mathbf{r}^T\left(\mathbf{n}_T''(0)+\mathbf{n}_{\mathbf{R}}''(0)\right)\xi^2. \tag{42}$$

For SAR systems, usually, we have $f_r \ll f_c$, so the following approximations can be made:

$$\begin{cases} \dfrac{f_c}{f_c+f_r} \approx 1-\dfrac{f_r}{f_c} \\ \left(\dfrac{f_c}{f_c+f_r}\right)^2 \approx 1-2\dfrac{f_r}{f_c}. \end{cases} \tag{43}$$

Substituting (43) into (41) and (42) yields

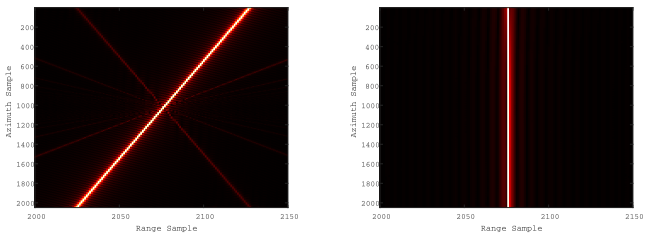
$$\Phi_{tr}(\xi, f_r) \approx \frac{2\pi R_0}{c}(f_r+f_c) + \frac{2\pi a_1' f_c}{c}\xi + \frac{2\pi a_2}{c}(f_c-f_r)\xi^2 + \frac{2\pi a_3}{c}(f_c-2\ f_r)\xi^3 \tag{44}$$

$$\Phi_{ro}(\xi, f_r) \approx \frac{2\pi(f_r+f_c)}{c}\mathbf{r}^T(\mathbf{n}_T(0)+\mathbf{n}_{\text{R}}(0)) + \frac{2\pi f_c}{c}\mathbf{r}^T\left(\mathbf{n}_T'(0)+\mathbf{n}_{\mathbf{R}}'(0)\right)\xi + \frac{\pi(f_c-f_r)}{c}\mathbf{r}^T\left(\mathbf{n}_T''(0)+\mathbf{n}_{\mathbf{R}}''(0)\right)\xi^2. \tag{45}$$

From (44) and (45), it can be seen that ξ and f_r terms are decoupled. Thus, the range walk has been eliminated.

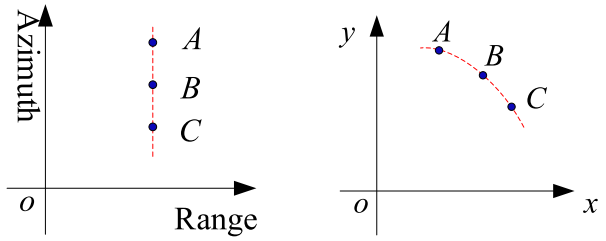
As shown in Figs. 5 and 6, due to the configuration of the BFSAR and the movement of the ship target, the received echo exhibits Doppler ambiguity. If the Doppler ambiguity has not been eliminated, the KT cannot correct the range

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(a) (b)

Fig. 6. KT processing results. (a) KT before Doppler ambiguity elimination. (b) KT after Doppler ambiguity elimination.



(a) (b)

Fig. 7. Range alignment diagram. (a) Before KT range alignment. (b) After KT range alignment.

migration. The processing result is shown in Fig. 6(a). From the result, it can be found that the RCM of the target energy trajectory still exists. In comparison, the KT result after Doppler ambiguity elimination is shown in Fig. 6(b). It is obvious that KT can correctly remove the range walk after Doppler ambiguity elimination. Therefore, Doppler deblurring is an essential process that must be completed in advance to correct the migration of ship targets. Meanwhile, this result also can illustrate the effectiveness of the Doppler ambiguity estimation method proposed in this article. The KT processed result is shown in Fig. 7, where we can find that these targets with the same initial bistatic range $R(0)$) are shifted to the same range gate after the KT range alignment.

After KT is utilized to eliminate the effect of the range walk, there may be residual high-order term migrations in the echo. However, in the study of ISAR imaging of ship targets, the accumulation time of azimuth imaging is relatively short, and the spatial variant of the residual high-order terms can be ignored. According to the previous analysis of echo property in Section II-C, range walk is the major component of RCM in BFSAR, and the higher-order terms are relatively small. Therefore, it is reasonable to ignore the spatial variant of the higher order terms in range alignment. Thus, the GRA method can be used for the range alignment of higher order terms.

After the envelope is aligned, phase errors caused by translation motion defocus the ISAR image, and it should be compensated for. In this article, the fast minimum entropy phase compensation (FMEPC) [41] method is applied to compensate for the translation phase. The image after phase compensation can be expressed as

$$g(k,n)=\sum_{m=0}^{M-1}h(m,n)\exp(-j\varphi_m)\exp\left(-j\frac{2\pi}{M}\text{km}\right) \quad (46)$$

where $h(m,n)$ is the aligned range profiles, $g(k,n)$ is the ISAR image, and φ represents the phase error to be compensated. k, m, and n represent the Doppler frequency, azimuth time, and range time.

Entropy can be used to measure the focusing quality of an image, and it is defined as

$$H_3=-\sum_{k=0}^{M-1}\sum_{n=0}^{N-1}\frac{|g(k,n)|^2}{S}\ln\frac{|g(k,n)|^2}{S} \quad (47)$$

where $S=\sum_{k=0}^{M-1}\sum_{n=0}^{N-1}|g(k,n)|^2$.

The minimum entropy criterion is applied to solve the error phase, so the error phase can be expressed as

$$\hat{\varphi}_m=\arg\left\{\min_{\varphi_m}[H_3]\right\},\quad m=0,1,\ldots,M-1. \quad (48)$$

When entropy reaches the minimum value, its corresponding gradient value is 0

$$\left.\frac{\partial H_3}{\partial\varphi_m}\right|_{\varphi_m=\hat{\varphi}_m}=0. \quad (49)$$

Through the solution formula (49), the phase value can be expressed as

$$\varphi_m=\text{angle}\{w^*(m)/|w(m)|\} \quad (50)$$

$$w(m)=\sum_{n=0}^{N-1}h(m,n)G(m,n) \quad (51)$$

$$G(m,n)=\text{FFT}\{[1+2\ln|g(k,n)|]\cdot g^*(k,n)\}. \quad (52)$$

The translation phase can be compensated by performing multiple iterations of formulas (50)–(52).

D. Joint FrFT-WVD Processing

As mentioned above, the range alignment is shown in Fig. 7; these targets with the same $R(0)$ are in the same range gate after the range alignment. Based on the analysis of the chirp rate in Section II-C, the chirp rate is the spatial variant. Thus, the targets in the same range gate have different chirp rates. In the turntable model of ISAR imaging, it is considered that all the scatterers have the same translational range history. Therefore, in the translational phase compensation, it is considered that all the scatterers have the same translational phase, so, after the translational phase compensation, the scatterers located in the same range gate still have a chirp rate, and the chirp rate of each scatterer is different. It is precise because of the spatial variant chirp rate that the RD imaging method based on the traditional FT will cause the image to be defocused. To address this issue, an azimuth focusing RID method based on joint FrFT-WVD is proposed.

Through the previous analysis, after translational compensation, the echo of a single range gate can be modeled as a superposition of multicomponent chirp signal

$$s(\eta)=\sum_{i=1}^{K}g_i(\eta)=\sum_{i=1}^{K}\sigma_i\exp\left\{j2\pi\left(\varphi_i+\alpha_i\eta+\frac{1}{2}\beta_i\eta^2\right)\right\} \quad (53)$$

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where σ_i, $2\pi\varphi_i$, α_i, and β_i are the amplitude, initial phase term, frequency, and chirp rate of the ith component, respectively. $i = 1, \ldots, K$, and K is the number of the component.

In this article, the aggregation characteristics of the chirp signal in the FrFT domain are used to realize the decomposition of the signal. The FrFT [42]–[44] of the LFM signal $s(\eta)$ with a rotation angle $\phi = p\pi/2$ and a fractional order p is defined by

$$S_p(u) = F^p[s(\eta)] = \int_{-\infty}^{\infty} s(\eta) K_p(\eta, u) d\eta \tag{54}$$

where $S_p(u)$ is the multicomponent signal in fractional domain, $F^p[\cdot]$ is the fractional transform with order p, and $K_p(\eta, u)$ is the kernel function of the FrFT, and it is given by

$$K_p(\eta, u) = \begin{cases} A_\phi e^{j\pi(\eta^2 \cot\phi - 2u\eta \csc\phi + u^2 \cot\phi)}, & \phi \neq n\pi \\ \delta(\eta - u), & \phi = 2n\pi \\ \delta(\eta + u), & \phi = (2n \pm 1)\pi \end{cases} \tag{55}$$

where $A_\phi = (1 - j\cot\phi)^{1/2}$ is the complex amplitude factor and n represents an integer.

Based on the property of the FrFT, a single chirp signal $g_i(\eta)$ exists an optimal rotation angle ϕ_i, which maximizes the integral amplitude of $S_p(u)$ after FrFT transformation. Correspondingly, $p_i = 2\phi_i/\pi$ is the optimal fractional order. At the optimal rotation angle, $S_p(u)$ will reach the peak at u_i. Based on this property, the optimal fraction order p_i and u_i can be obtained by

$$\{p_i, u_i\} = \arg\max_{p,u} \left| S_p(u) \right|. \tag{56}$$

FrFT and FFT are both linear operators [42]–[44]. The multicomponent signal of different frequencies can be separated by a narrow frequency filter. Similarly, the multicomponent chirp signal can also be separated by a narrow frequency filter in the fractional domain

$$g_i(\eta) = F^{-p_i}\left[S_p(u) \mathrm{win}(u_i) \right] \tag{57}$$

where $F^{-p}[\cdot]$ is the inverse fractional transform with order p and $\mathrm{win}(u_i)$ is the narrow window of the filter around the strongest peak point in the fractional domain associated with u_i. After the inverse fractional transform, the time-domain signal of the strongest signal component $g_i(\eta)$ can be obtained. Find the first ascending point from u_i to the left and to the right, and denote it as u_l and u_r, respectively. Then, the window can be expressed as

$$\mathrm{win}(u_i) = \mathrm{rect}\left[\frac{u - u_{\mathrm{cen}}}{L} \right] \tag{58}$$

where $L = u_r - u_l$ and $u_{\mathrm{cen}} = (u_r + u_l)/2$ are the length and center of the rectangular window, respectively.

After separating the signal $g_i(\eta)$ from $s(\eta)$, the WVD can be calculated to obtain a high-resolution time–frequency distribution, and the definition of WVD is given by

$$w(\eta, f) = \mathrm{WVD}[s(\eta)] = \int s\left(\eta + \frac{\tau}{2}\right) s^*\left(\eta - \frac{\tau}{2}\right) e^{-j2\pi\tau} d\tau. \tag{59}$$

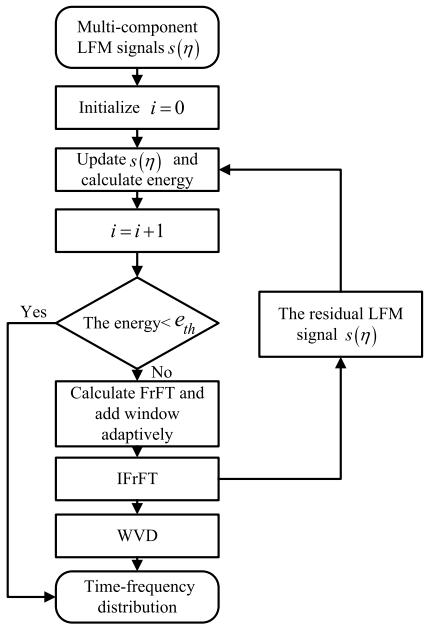


Fig. 8. Flowchart of the time–frequency distribution multicomponent LFM signal via joint FrFT-WVD processing.

TABLE I
PARAMETERS OF THE THREE-COMPONENT CHIRP SIGNALS

Components (i)	σ_i	$\varphi_i(rad)$	α_i	β_i
1	1	$\pi/6$	-5	-3
2	1	$\pi/4$	5	3
3	1.2	$-\pi/6$	10	6

Then, the Doppler frequency at any instantaneous moment can be obtained. The flowchart of the time–frequency distribution multicomponent LFM signal via joint FrFT-WVD processing is shown in Fig. 8.

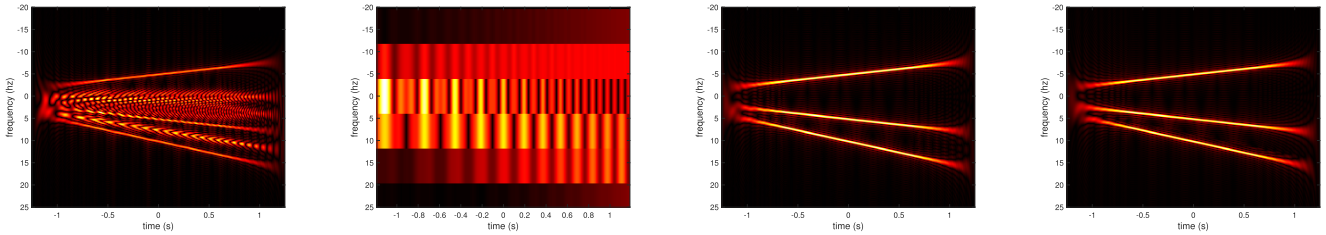
Consider a three-component chirp signal as in (53), where the parameters are given in Table I. From Fig. 9(a) and (b), the WVD distribution has a high-resolution time–frequency distribution but has serious cross terms. Although STFT does not have cross terms, according to the Heisenberg time–frequency uncertainty principle, the time–frequency resolution is contradictory, and the resolution is not high. Comparing Fig. 9(d) with (c), it can be seen from Fig. 10 that the method proposed in this article can achieve the effect of calculating WVD separately for each signal. Not only can the cross term be perfectly suppressed but there is no loss in time–frequency resolution.

IV. MAIN STEPS OF THE PROPOSED METHOD

Up until now, the hybrid SAR-ISAR image formation via the joint FrFT-WVD has been analyzed in detail. Its flowchart is demonstrated in Fig. 11, and the main steps are listed as follows.

Step 1: After range compression, the minimum entropy criterion of the ARP is applied to estimate the Doppler center ambiguity.

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(a) (b) (c) (d)

Fig. 9. Time–frequency distribution results of three-signal components. (a) WVD. (b) STFT. (c) WVD for each signal separately. (d) Joint FrFT-WVD.

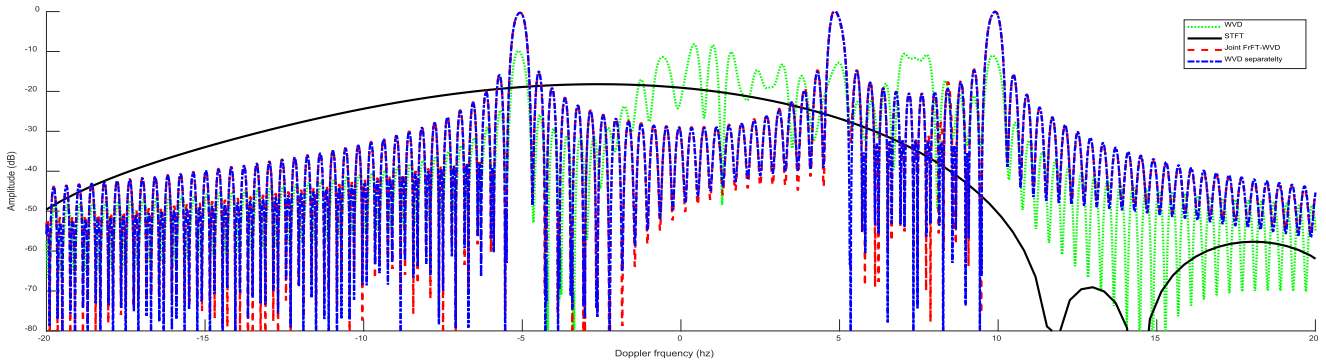


Fig. 10. Doppler profile comparison of different methods at 0 s.

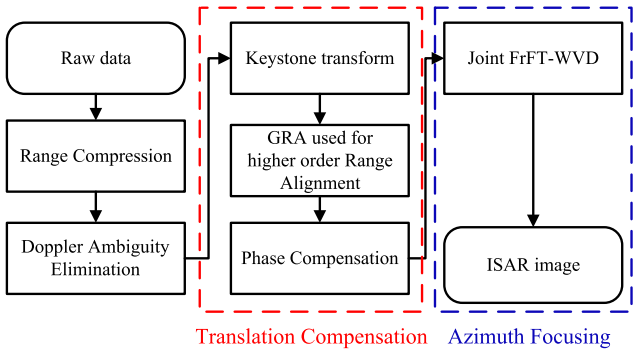


Fig. 11. Flowchart of the proposed hybrid SAR-ISAR image formation.

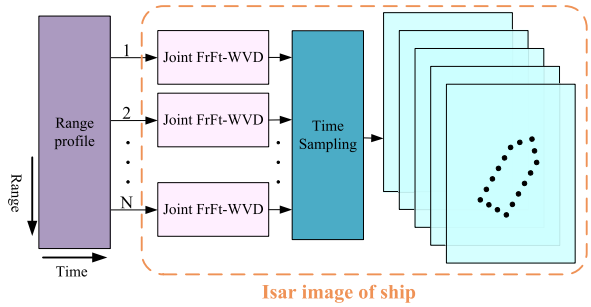


Fig. 12. Flowchart of the proposed RID imaging algorithm.

Step 2: Remove Doppler centroid ambiguity based on estimated parameter.

Step 3: KT is applied to correct the range walk, and then, the minimum entropy criterion of the ARP is used to align high-order terms. After range alignment, compensate for the translation phase.

Step 4: Resolve the Doppler ambiguity and translate motion compensation. Obtain the echo of the n_rth range cell $s(\eta)$.

Step 5: Initialize $n_r = 1$ and $i = 1$.

Step 6: Calculate the FrFT transform of the signal $s(\eta)$.

Step 7: Find the peak value of $S_p(u)$, which means that a component with the strongest energy is selected. Then, acquire p_i, u_i based on (56) from the peak.

Step 8: Construct a narrow window according to u_i, and filter the adaptive fractional domain to obtain the signal $g_i(\eta)$ based on (57).

Step 9: Calculate the WVD of the signal $g_i(\eta)$, and sum it up

$$\begin{cases} w_i(\eta, f_\eta) = \mathrm{WVD}(g_i(\eta)) \\ w_{\mathrm{sum}}(\eta, f_\eta) = w_{\mathrm{sum}}(\eta, f_\eta) + w_i(\eta, f_\eta). \end{cases} \tag{60}$$

Step 10: Eliminate the ith component. Then, update $i = i + 1$.

Step 11: Repeat Steps 6–10 until the energy of $s(\eta)$ is below a threshold e_{th} (usually 0.1 of the initial signal). Then, update $n_r = n_r + 1$.

Step 12: Repeat Steps 6–11 until each range bin has been processed ($n_r = N$), where N represents the range sampling number.

After all range bins have been processed by the method aforementioned, the time–frequency distribution of each range gate can be obtained. Once the moment is chosen, a well-focused ship target image can be reconstructed. Fig. 12 shows the flowchart of the proposed RID imaging algorithm.

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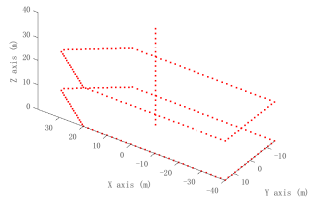


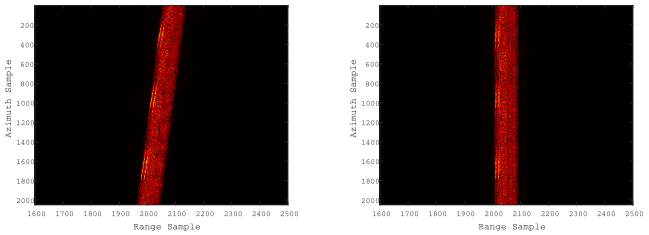
Fig. 13. Scatterer model of the ship target.

TABLE II
BFSAR SYSTEM PARAMETERS

Parameter	Symbols name	Values
Carrier Frequency	f_c	9.6Ghz
Range Bandwidth	B_r	200Mhz
Pulse Repetition Frequency	f_r	2000Hz
Receiver's Position	(X_R, X_R, H_R)	(0,-8000,3000)m
Transmitter's Position	(X_T, X_T, H_T)	(4000,-2600,2000)m
Platform Velocity	V	77m/s

TABLE III
ROTATION PARAMETERS

Rotation Parameters	Amplitude (deg)	Period (s)	Initial phase (rad)
Roll	4	14	0
Pitch	5	7	$\pi/9$
Yaw	3.6	15	$\pi/12$



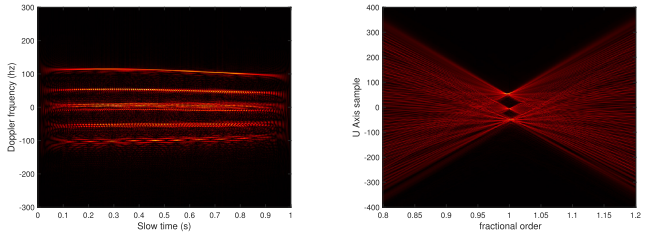
(a) (b)

Fig. 14. Simulated data processing results. (a) Range compression result. (b) Translational compensation result.

V. SIMULATION RESULTS

To verify the effectiveness of the proposed joint FrFT-WVD RID imaging method, simulations are carried out in this section, and the imaging results based on RD, STFT, and WVD are also provided for comparison. The scatterer model of the simulated extended target is shown in Fig. 13. In this simulated scenario, the receiver and transmitter aircraft platforms fly along the Y-axis, and both the velocities are $V = 77$ m/s. The original coordinates of receiver and transmitter are $(0, -8000, 3000)$ and $(4000, -2600, 2000)$ m. The detailed BFSAR system parameters are given in Table II. The rotation parameters of ship target are listed in Table III. The center of the ship is located at $(0\ \text{m}, 0\ \text{m}, 0\ \text{m})$, and the ship moves with a velocity of $(7\ \text{m/s}, 11\ \text{m/s}, 0\ \text{m/s})$.

In the following, a noise-free example is given to verify the effectiveness of the method proposed in this article. Fig. 14(a) shows the received echo data of the ship target in the range-compressed domain. It can be obviously seen that the echo signal is dispersed into multiple range gates, which means that the echo of the BFSAR exits RCM. Fig. 14(b) shows the trajectory of the simulated ship after translational compensation. Comparing Fig. 14(a) with (b), the energy trajectory of the scatterers changes from slant to vertical, which means that the RCM of the BFSAR has been corrected to a great extent. Fig. 15(a) shows the WVD of the BFSAR echo from the 2081st range cell. It can be observed that the curvature of the distribution is very small, which means that the third-order Doppler parameter of the echo signal can be ignored after translational compensation. It can also be found from Fig. 15(b) that the echo energy can be gathered in the FrFT domain to form multiple peak points. Therefore, it is reasonable to model the echo signal of each range gate as a multicomponent chirp signal. Meanwhile, there are many cross terms in Fig. 15(a), so it is not appropriate to use WVD imaging directly.



(a) (b)

Fig. 15. One range gate processing result. (a) WVD result. (b) FrFT result.

TABLE IV
ENTROPIES OF IMAGES

	RD	STFT	WVD	joint FrFT-WVD
Entropies	8.1175	7.7477	8.4350	6.6605

Fig. 16 shows the image results of the reconstruction of the scatterers. The imaging result of the traditional RD algorithm is shown in Fig. 16(a). The image is seriously defocused, and the resolution is low. The RID imaging result based on STFT is shown in Fig. 16(b). The resolution of the imaging result is improved compared with RD, but the resolution is still low. As shown in Fig. 16(c), the imaging result of WVD has a high resolution, but there are many false objects in the image because of the existence of cross terms. In general, the proposed joint FrFT-WVD-based imaging RID method can reconstruct the scatterers correctly and has high processing accuracy compared to other methods.

To evaluate the quality of the image, the entropies of these images are also calculated. The entropy value can measure the focusing performance of the image. The ISAR image has the best focusing performance, and the recovered scattering object has a clear structure when the entropy value is the minimum. Table IV shows the calculation results of image entropy. Results in Table IV indicate that the image reconstructed by the joint FrFT-WVD method has the best image quality.

In order to verify the performance of the algorithm proposed in this article, random noise has also been added to the echo, and the SNR is set to be -10 dB. As shown in Fig. 17, the proposed joint FrFT-WVD method can still reconstruct the structure of the scatterer correctly, and the image is well focused. However, the image becomes more defocused for other imaging methods. Meanwhile, from Table V, it can be

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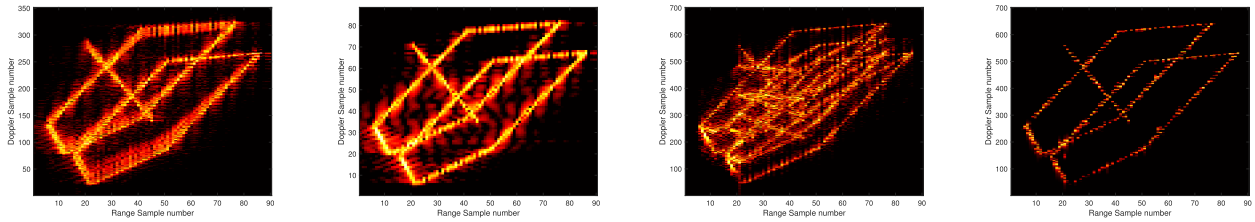


Fig. 16. Image results of simulated data in noise free condition. (a) RD. (b) STFT. (c) WVD. (d) Joint FrFT-WVD.

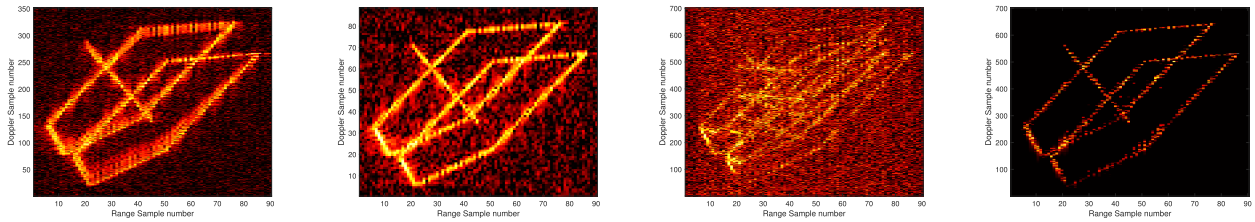


Fig. 17. Image results of simulated data with received SNR $= -10$ dB. (a) RD. (b) STFT. (c) WVD. (d) Joint FrFT-WVD.

TABLE V
SNR OF IMAGE RESULTS

	RD	STFT	WVD	joint FrFT-WVD
SNR (dB)	32.3829	32.0667	14.7488	53.9622

noticed that the joint FrFT-WVD method can obtain a higher SNR. In summary, the proposed method shows a stronger antinoise performance compared with the other methods.

VI. CONCLUSION

This article proposes a kind of hybrid SAR-ISAR imaging formation for BFSAR ship target imaging. First, to solve the problem of the Doppler ambiguity caused by the forward-looking mode of the receiver, an efficient ambiguity estimation method based on the minimum entropy criterion is presented. Then, KT and range alignment processing can be applied to correct the spatial variant range walk and spatial invariant higher order RCM, respectively. Finally, after the translational compensation, joint FrFT-WVD processing is carried out for each range gate to obtain a well-focused ISAR image. The simulation results have verified the effectiveness of the proposed method, i.e., it is effective and suitable for BFSAR imaging of ship targets. The main advantages of this method are not only the absence of cross terms but also the high processing accuracy and better antinoise performance.

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